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ASSIGNMENT – 3

Execute Different Commands of Hadoop on a Single Node Cluster

Below is the step-by-step execution of fundamental Hadoop Distributed File System (HDFS) commands.

Command 1: Create a Directory (mkdir) and List Contents (ls)
Created a new directory named /assignment3 in the HDFS root and verified its creation.

```
kanishka@master:~$ hdfs dfs -mkdir /assignment3_kanishka
kanishka@master:~$ hdfs dfs -ls /
Found 1 items
drwxr-xr-x   - kanishka supergroup          0 2026-04-22 10:23 /assignment3_kanishka
kanishka@master:~$
```

Command 2: Upload a Local File to HDFS (put)
Created a local text file and uploaded it to the /assignment3 directory using the -put command.

```
kanishka@master:~$ echo "Hello Hadoop Assignment" > file.txt
kanishka@master:~$ hdfs dfs -put file.txt /assignment3_kanishka
kanishka@master:~$ hdfs dfs -ls /assignment3
ls: `/assignment3': No such file or directory
kanishka@master:~$ hdfs dfs -ls /assignment3_kanishka
Found 1 items
-rw-r--r--   1 kanishka supergroup         24 2026-04-22 10:24 /assignment3_kanishka/file.txt
kanishka@master:~$
```

Command 3: Read a File in HDFS (cat)
Displayed the contents of the uploaded text file directly from HDFS using the -cat command.

```
kanishka@master:~$ hdfs dfs -cat /assignment3_kanishka/file.txt
Hello Hadoop Assignment
```

Command 4: Download a File to Local System (get)
Downloaded the file from HDFS back to the local Ubuntu file system using the -get command.

```
kanishka@master:~$ hdfs dfs -get /assignment3_kanishka/file.txt downloaded.txt
kanishka@master:~$ ls
downloaded.txt  file.txt  hadoop-3.3.6.tar.gz
kanishka@master:~$
```

Command 5: Delete a File from HDFS (rm)
Removed the file from the HDFS directory to clean up the workspace using the -rm command.

```
kanishka@master:~$ hdfs dfs -rm /assignment3_kanishka/file.txt
Deleted /assignment3_kanishka/file.txt
```

```
kanishka@master:~$ hdfs dfs -ls /assignment3_kanishka
kanishka@master:~$
```